

Technology Stack

Date	10 Jul 2026
Team ID	xxxxxx
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to build a responsive, interactive, and user-friendly web interface for loan prediction and applicant management.

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Handles business logic, processes user requests, integrates the machine learning model, and generates prediction results efficiently.</i>
3	Database / Data Storage	<i>SQLite (Local Database)</i>	<i>Stores applicant details, prediction history, and loan records securely for quick retrieval and management.</i>
4	Cloud / Hosting / Deployment	<i>Local Host (Flask Development Server)</i>	<i>Used for project development, testing, and demonstration without requiring cloud deployment.</i>

5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Manages source code, tracks project changes, and enables version control and collaboration.</i>
6	Third-Party APIs / Other Tools	<i>Scikit-learn, Pandas, NumPy, XGBoost, Joblib, Chart.js</i>	<i>Used for data preprocessing, machine learning model training, prediction generation, model loading, and visualization of analytics.</i>